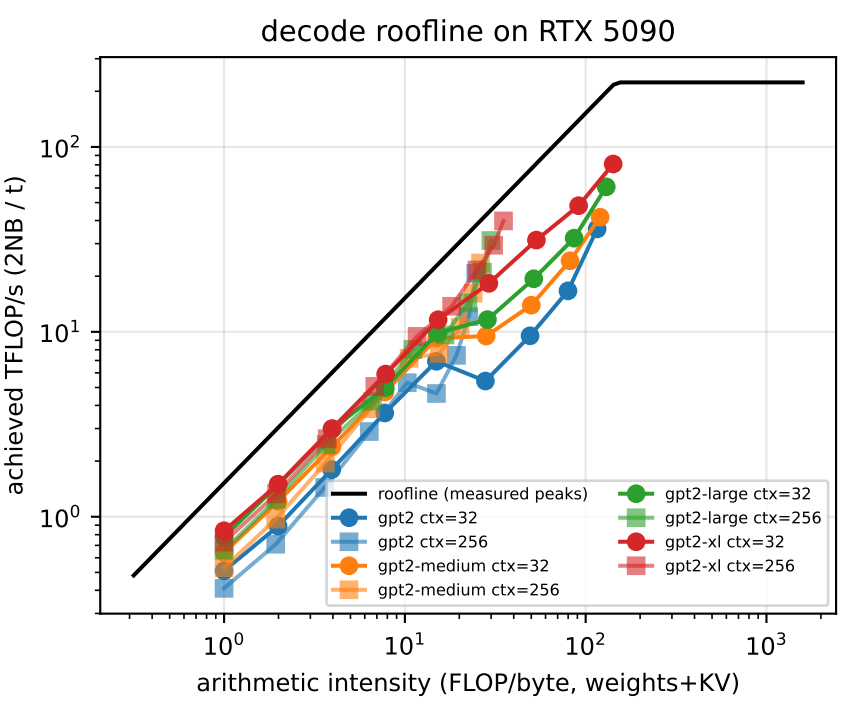
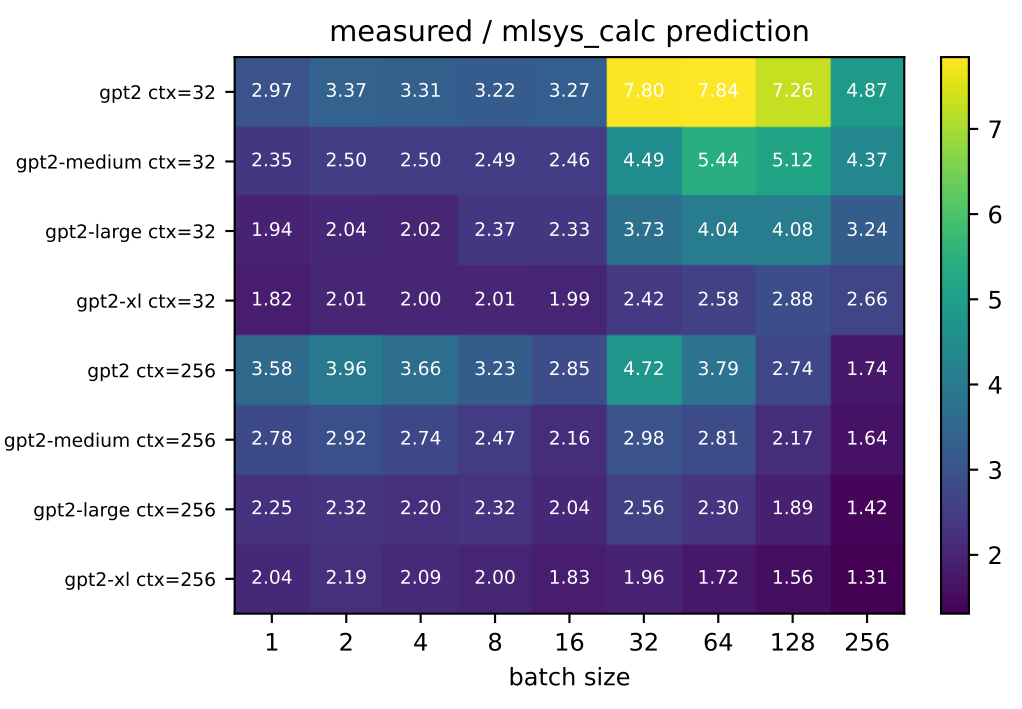

decode roofline on RTX 5090
achieved TFLOP/s (2NB / t)
arithmetic intensity (FLOP/byte, weights+KV)
roofline (measured peaks)
gpt2 ctx=32
gpt2 ctx=256
gpt2-medium ctx=32
gpt2-medium ctx=256
gpt2-large ctx=32
gpt2-large ctx=256
gpt2-xl ctx=32
gpt2-xl ctx=256


measured / mlsys_calc prediction
gpt2 ctx=32 2.97 3.37 3.31 3.22 3.27 7.80 7.84 7.26 4.87
gpt2-medium ctx=32 2.35 2.50 2.50 2.49 2.46 4.49 5.44 5.12 4.37
gpt2-large ctx=32 1.94 2.04 2.02 2.37 2.33 3.73 4.04 4.08 3.24
gpt2-xl ctx=32 1.82 2.01 2.00 2.01 1.99 2.42 2.58 2.88 2.66
gpt2 ctx=256 3.58 3.96 3.66 3.23 2.85 4.72 3.79 2.74 1.74
gpt2-medium ctx=256 2.78 2.92 2.74 2.47 2.16 2.98 2.81 2.17 1.64
gpt2-large ctx=256 2.25 2.32 2.20 2.32 2.04 2.56 2.30 1.89 1.42
gpt2-xl ctx=256 2.04 2.19 2.09 2.00 1.83 1.96 1.72 1.56 1.31
1 2 4 8 16 32 64 128 256
batch size